Investigate Newton’s laws of motion and quantitatively analyse the relationship between force, mass and acceleration of objects

Learning intention

- investigate Newton’s laws of motion and quantitatively analyse the relationship between force, mass and acceleration of objects.

Teach and model

- investigating how driverless vehicles apply Newton’s laws of motion to brake in time.
- investigating a moving object to analyse and propose relationships between distance and time, speed, force and acceleration.
- constructing an argument, supported by data, to support lower speed limits near schools or for trucks in urban environments.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.